

ECON4200 - Hong Kong Statutory Minimum Wage Analysis - Age Group

2026-04-24

setting up again

Load Libraries & Define Mappings

load lib

done load lib

load earning map

```
earnings_map <- tibble(  
  code = str_pad(1:40, width = 2, pad = "0"),  
  midpoint_hkd = c(1000,2500,3500,4500,5500,6500,7500,8500,9500,  
                   10500,11500,12500,13500,14500,15500,16500,17500,  
                   18500,19500,21000,23000,25000,27000,29000,31000,  
                   33000,35000,37000,39000,41000,43000,45000,47000,  
                   49000,55000,65000,75000,85000,95000,120000)  
)  
  
other_earnings_map <- tibble(  
  code = str_pad(1:11, width = 2, pad = "0"),  
  midpoint_hkd = c(500,1500,2500,3500,4500,5500,6500,7500,8500,9500,12000)  
)
```

done load earning map

Data Loading & Processing

Load all files + calculate est hourly wage same method used

done

ranges mapping

done

CPI Processing (Real Wages)

```
## [1] "2011 Q2 Reference CPI Value is: 78"
```

done

Add Real Wage to Main Dataset

done adding real wage

Testing to reconfirm is same as previous rmd file

```
## === Basic Data Overview ===

## Total rows in raw data: 1300115

## Total rows after processing: 1300115

## Rows in focus sample: 572003

## Years covered:

## # A tibble: 112 x 3
##   year quarter    n
##   <dbl>   <dbl> <int>
## 1  1985         1  4502
## 2  1985         2  4807
## 3  1985         3  4698
## 4  1985         4  4769
## 5  1986         1  4732
## 6  1986         2  4832
## 7  1986         3  4659
## 8  1986         4  4683
## 9  1987         1  4627
## 10 1987         2  4743
## # i 102 more rows

##

## === Key Variable Summary ===

##   hourly_wage    real_hourly_wage    weight    hours_main_employment
##   Min.      : 2.325   Min.      : 2.219   Min.      : 0.0003   Min.      : 8.00
##   1st Qu.: 27.170   1st Qu.: 37.459   1st Qu.: 0.0005   1st Qu.:40.00
##   Median : 45.208   Median : 54.929   Median : 0.0005   Median :48.00
##   Mean   : 69.905   Mean   : 83.719   Mean   :142.7027   Mean   :46.78
##   3rd Qu.: 79.114   3rd Qu.: 91.772   3rd Qu.: 0.0008   3rd Qu.:54.00
##   Max.    :3452.244   Max.    :4090.254   Max.    :775.6073   Max.    :99.00
## earnings_main_hkd
##   Min.      : 1000
##   1st Qu.: 5500
##   Median : 9500
##   Mean   : 13038
##   3rd Qu.: 15500
##   Max.    :120000

##

## === Missing Values (%) ===

## # A tibble: 19 x 2
##   Variable    `Missing_%`
##   <chr>      <dbl>
## 1 industry_underemployed    100
## 2 previous_industry          100
## 3 previous_occupation        100
## 4 unemployment_duration      100
## 5 leave_reason               100
## 6 intend_same_industry        100
## 7 intend_industry             100
## 8 intend_same_occupation      100
```

```
## 9 intend_occupation      100
## 10 hours_secondary      99.7
## 11 year_completed_study  94.5
## 12 parent_serial        73.3
## 13 length_employment    58.8
## 14 spouse_serial        56.6
## 15 work_place           53.8
## 16 relationship         24.0
## 17 foreign_helper       24.0
## 18 other_employment_7days 24.0
## 19 earnings_underemployed 1.84

##
## === Hourly Wage Distribution (Nominal) ===
```

Min	P5	Median	Mean	P95	Max	N
2.32	11.99	45.21	69.91	206.54	3452.24	572003

1. Check Earnings Mapping

```
## Unique earnings_main codes and their mapped values:
```

```
## # A tibble: 40 x 3
##   earnings_main earnings_main_hkd     n
##   <chr>          <dbl> <int>
## 1 01              1000 21222
## 2 02              2500 29452
## 3 03              3500 33208
## 4 04              4500 32707
## 5 05              5500 35875
## 6 06              6500 42288
## 7 07              7500 41219
## 8 08              8500 46906
## 9 09              9500 33480
## 10 10             10500 40181
## 11 11             11500 18784
## 12 12             12500 25518
## 13 13             13500 15513
## 14 14             14500  8794
## 15 15             15500 26255
## 16 16             16500  7180
## 17 17             17500  5605
## 18 18             18500  8614
## 19 19             19500  3158
## 20 20             21000 25134
## 21 21             23000  7187
## 22 22             25000 10608
## 23 23             27000  3379
## 24 24             29000  3022
## 25 25             31000 12607
## 26 26             33000  2149
## 27 27             35000  4402
## 28 28             37000  1147
## 29 29             39000  1011
```

```
## 30 30                                41000  5255
## # i 10 more rows
```

2. Check Weighting Factor

```
## Total weighted population (should be around Hong Kong's employed population):
```

```
## [1] 81626372
```

3. Sample of Raw vs Processed Data

yea this one isnt going to be same but uh it exist

year	quarter	earnings_main	earnings_main_hk	hours_main_employment	hourly_wage	real_hourly_wage
1986	2	02	2500	48	11.99	30.86
2003	2	15	15500	54	66.06	77.80
2009	2	07	7500	54	31.97	34.53
1987	1	01	1000	48	4.79	11.96
2009	2	09	9500	48	45.55	49.21
1987	2	04	4500	48	21.58	52.65
1994	2	08	8500	48	40.76	52.86
1999	1	12	12500	54	53.28	55.31
1997	4	22	25000	44	130.77	134.09
2010	1	19	19500	72	62.33	65.82

Now actual content

Cont. The aggregate data shows a continued improvement in wages and a reduction in the low-wage share around 2011, the effect appears modest in the overall population. This raises the question of whether the minimum wage had a stronger impact on specific vulnerable groups. Therefore, the next step is to conduct a more disaggregated analysis by examining subgroups such as:

Younger workers (e.g., aged 15–29) & prime workers maybe

Age grouping

Age code: Code Description 01 < 5 02 5 – 9 03 10 – 14 04 15 – 19 05 20 – 24 06 25 – 29 07 30 – 34 08 35 – 39 09 40 – 44 10 45 – 49 11 50 – 54 12 55 – 59 13 60 – 64 14 > 65

We decided to split it as

15-29 vs 30-49 vs 50-64

15-29 captures the critical transition period (students, fresh graduates, early career)

30-49 serves as a strong control group — they are less likely to be affected by the minimum wage

50-64 allows us to see whether older low-skilled workers also benefited (important in Hong Kong context with an aging population)

We group them separately:

grouping done yes i didnt set echo=true here so you cannot see the codes, i dont head(df_focus) cuz there are like 40+ variables or sth it is going to be a long printing.

Here's the % occupied for each age group:

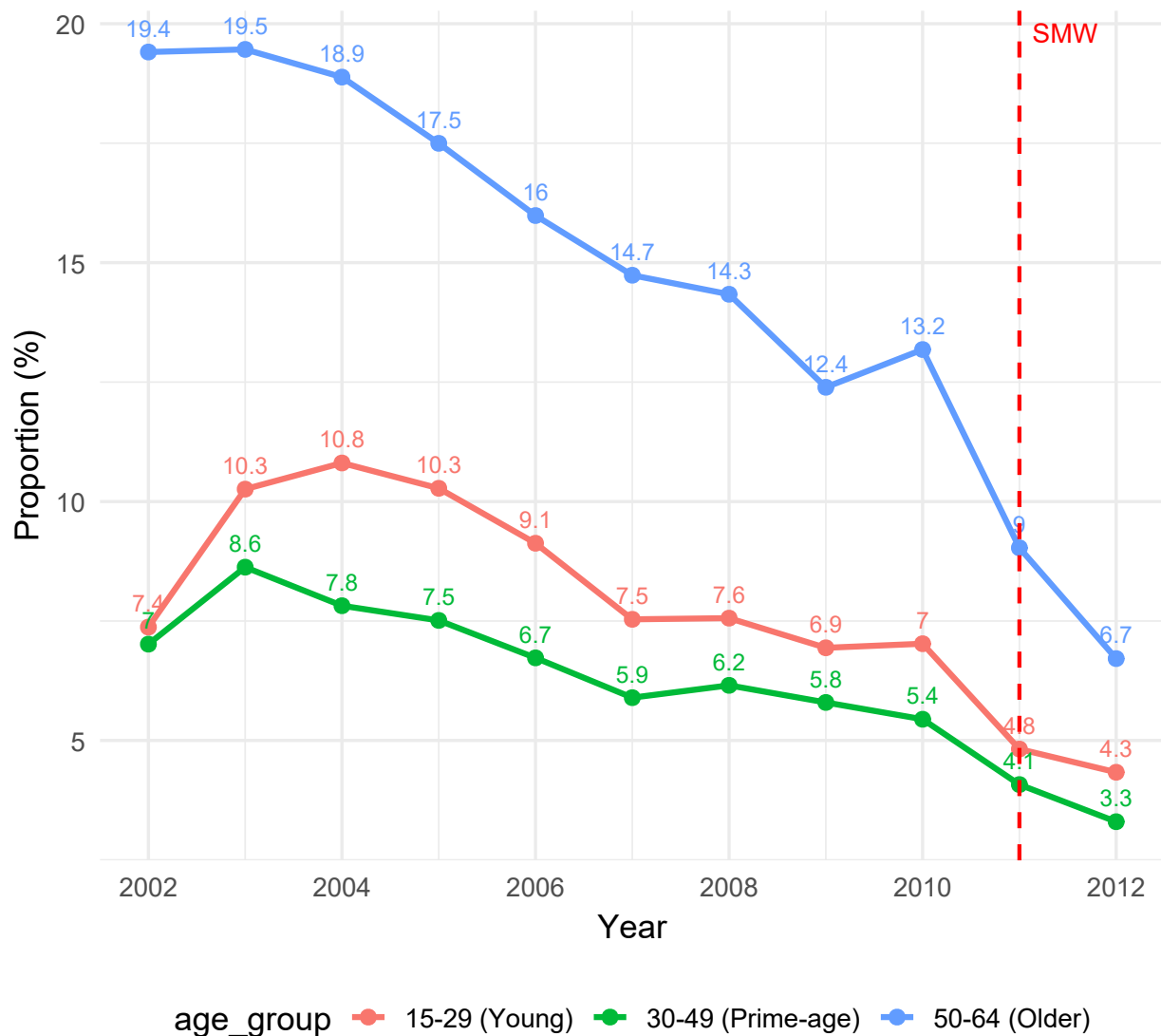
```
## # A tibble: 3 x 3
##   age_group      n    pct
##   <chr>      <int> <dbl>
## 1 15-29 (Young) 163226 28.5
## 2 30-49 (Prime-age) 306917 53.7
## 3 50-64 (Older) 101860 17.8
```

1. Long-term Trend (Proportion Below \$28 by Age Group)

```
## # A tibble: 3 x 3
##   year age_group    below_28
##   <dbl> <chr>      <dbl>
## 1 2008 50-64 (Older)    14.3
## 2 2010 50-64 (Older)    13.2
## 3 2012 50-64 (Older)     6.71
```

Proportion Earning Below Real HK\$28 by Age Group

Real wages in 2011 prices



YoY % Change Tables by Age Group

2008 to 2012, 2007 is included as yoy% change for 2008 else 2008 % change will be NA

```
##
## ## 15-29 (Young)
##
##
## Table: Proportion Below Thresholds & YoY % Change - 15-29 (Young)
##
## | year| Below $28| Below $35| Below $40| YoY $28| YoY $35| YoY $40|
## |----:|-----:|-----:|-----:|-----:|-----:|-----:|
## | 2008|      7.6|     18.3|     29.1|      1.3|      1.7|      1.4|
```

```

## | 2009|      6.9|      17.7|      27.5|     -9.2|     -3.3|     -5.5|
## | 2010|      7.0|      18.8|      27.7|      1.4|      6.2|      0.7|
## | 2011|      4.8|      14.9|      25.9|    -31.4|    -20.7|     -6.5|
## | 2012|      4.3|      14.1|      24.5|    -10.4|     -5.4|     -5.4|
##
##
## ## 30-49 (Prime-age)
##
##
##
## Table: Proportion Below Thresholds & YoY % Change - 30-49 (Prime-age)
##
## | year| Below $28| Below $35| Below $40| YoY $28| YoY $35| YoY $40|
## |----:|-----:|-----:|-----:|-----:|-----:|-----:|
## | 2008|      6.2|      12.8|      18.0|      5.1|      7.6|      5.9|
## | 2009|      5.8|      13.0|      18.5|     -6.5|      1.6|      2.8|
## | 2010|      5.4|      11.7|      16.8|     -6.9|    -10.0|     -9.2|
## | 2011|      4.1|      10.3|      15.6|    -24.1|    -12.0|     -7.1|
## | 2012|      3.3|       9.8|      15.2|    -19.5|     -4.9|     -2.6|
##
##
## ## 50-64 (Older)
##
##
##
## Table: Proportion Below Thresholds & YoY % Change - 50-64 (Older)
##
## | year| Below $28| Below $35| Below $40| YoY $28| YoY $35| YoY $40|
## |----:|-----:|-----:|-----:|-----:|-----:|-----:|
## | 2008|     14.3|     27.4|     34.3|     -2.7|      5.0|      3.3|
## | 2009|     12.4|     24.9|     32.6|    -13.3|     -9.1|    -5.0|
## | 2010|     13.2|     23.8|     30.8|      6.5|     -4.4|    -5.5|
## | 2011|      9.0|     21.2|     29.5|    -31.8|    -10.9|    -4.2|
## | 2012|      6.7|     19.9|     27.9|    -25.6|     -6.1|    -5.4|
##
## # A tibble: 18 x 6
##   year age_group `Below $28` `Below $35` `Below $40`   n
##   <dbl> <chr>      <dbl>      <dbl>      <dbl> <int>
## 1  2007 15-29 (Young)      7.54      18.0      28.7  5127
## 2  2008 15-29 (Young)      7.56      18.3      29.1  5000
## 3  2009 15-29 (Young)      6.94      17.7      27.5  4846
## 4  2010 15-29 (Young)      7.02      18.8      27.7  4485
## 5  2011 15-29 (Young)      4.82      14.9      25.9  4424
## 6  2012 15-29 (Young)      4.33      14.1      24.5  4420
## 7  2007 30-49 (Prime-age)   5.90      11.9      17.0 12326
## 8  2008 30-49 (Prime-age)   6.15      12.8      18.0 11887
## 9  2009 30-49 (Prime-age)   5.80      13.0      18.5 11560
## 10 2010 30-49 (Prime-age)   5.44      11.7      16.8 11234
## 11 2011 30-49 (Prime-age)   4.08      10.3      15.6 11081
## 12 2012 30-49 (Prime-age)   3.30       9.76      15.2 10775
## 13 2007 50-64 (Older)     14.7      26.1      33.2  4815
## 14 2008 50-64 (Older)     14.3      27.4      34.3  5170
## 15 2009 50-64 (Older)     12.4      24.9      32.6  5315
## 16 2010 50-64 (Older)     13.2      23.8      30.8  5325

```

##	17	2011	50-64 (Older)	9.03	21.2	29.5	5737
##	18	2012	50-64 (Older)	6.71	19.9	27.9	5997

Key Observations:

Older workers (50-64) have by far the highest proportion earning below HK\$28 — both before (13%) and after (6.8%).

Young workers (15-29) improved, but started from a lower base.

Prime-age (30-49) have the lowest incidence of very low wages.

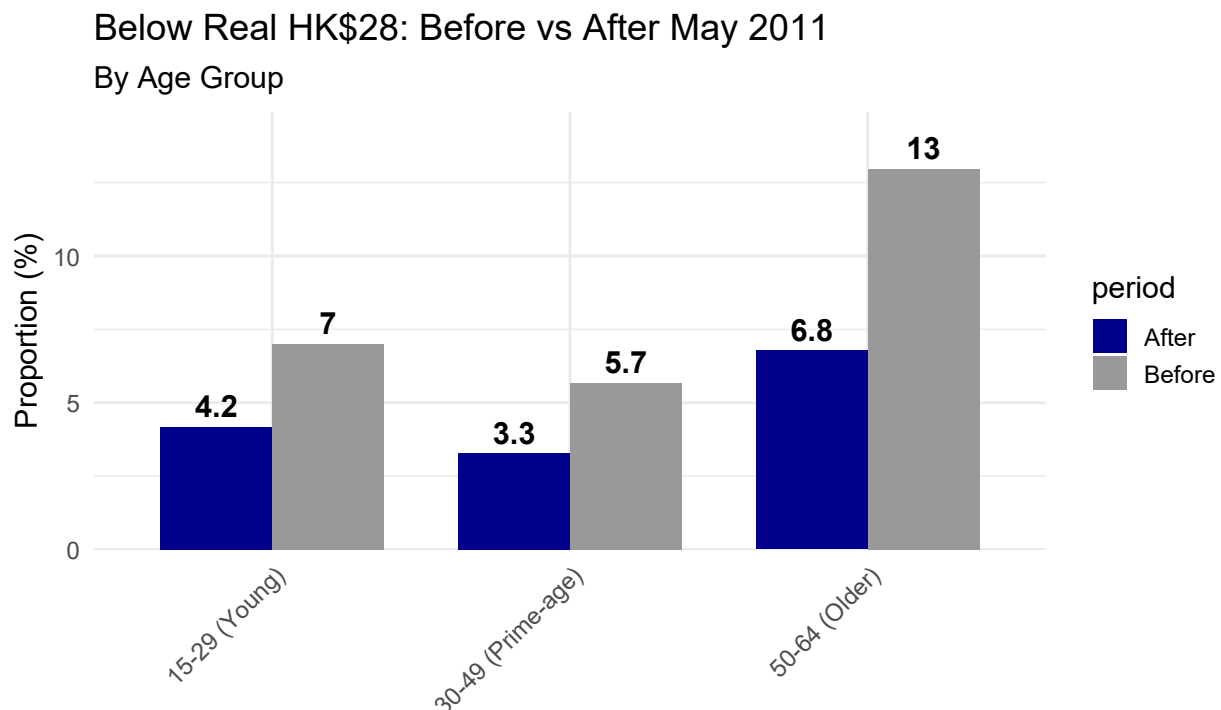
Possible reasons:

Older workers (50-64) in Hong Kong: Many are in low-paying, low-skill service jobs (cleaning, security, retail, restaurant helpers). They often have lower bargaining power, less formal education, and may face age discrimination. Some are re-entering the labor market after raising children or due to insufficient retirement savings. This group is surprisingly vulnerable in Hong Kong's labor market.

Young workers (15-29): Many are still studying or in part-time / training positions. Some are in higher-paying entry-level white-collar or service jobs. They benefited from the minimum wage, but the baseline was not as bad as for older low-skilled workers.

2. Before vs After by Age Group (Below \$28) - Main Plot

2008Q1 - 2011Q1 vs 2011Q2 - 2012Q4



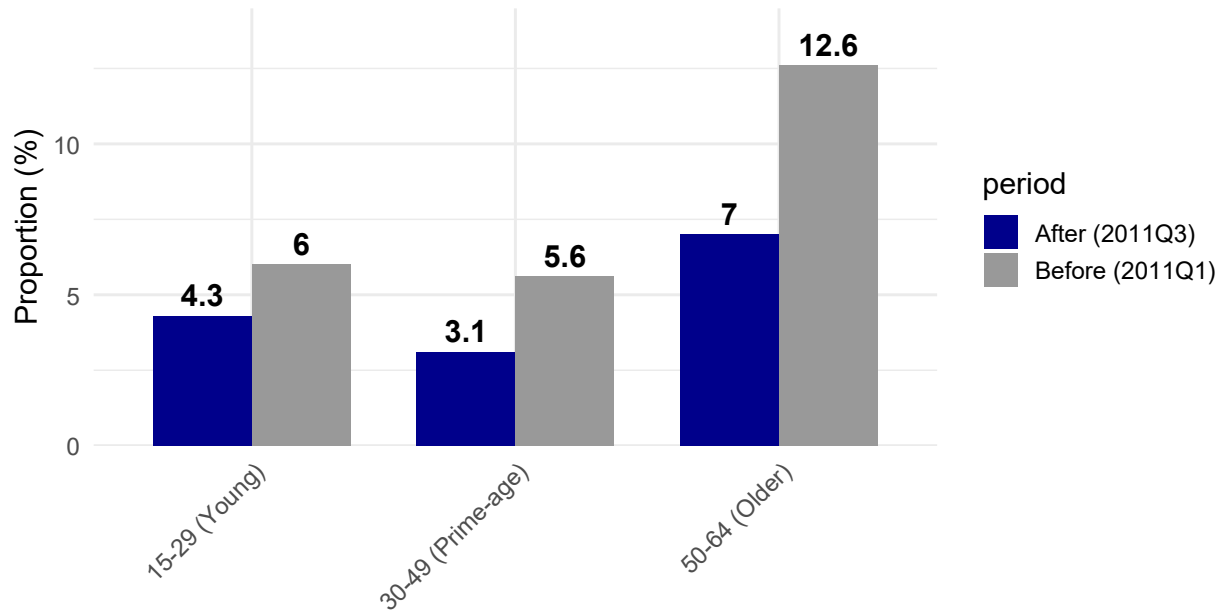
But then there could be averaging effect like 2008Q1 to 2011Q1 vs 2011Q2 - 2011Q4, so we dont use avg effect BUT we do right before and right after i.e. 2011Q1 vs 2011Q3 directly:

```
## # A tibble: 6 x 5
##   period    age_group `Below $28` `Below $35` `Below $40`
##   <chr>      <chr>      <dbl>      <dbl>      <dbl>
```


## 1	After (2011Q3)	15-29 (Young)	4.3	13	23.7
## 2	After (2011Q3)	30-49 (Prime-age)	3.1	9	13.8
## 3	After (2011Q3)	50-64 (Older)	7	20.7	28.4
## 4	Before (2011Q1)	15-29 (Young)	6	17.2	28.6
## 5	Before (2011Q1)	30-49 (Prime-age)	5.6	12.5	17.5
## 6	Before (2011Q1)	50-64 (Older)	12.6	23.1	30.5

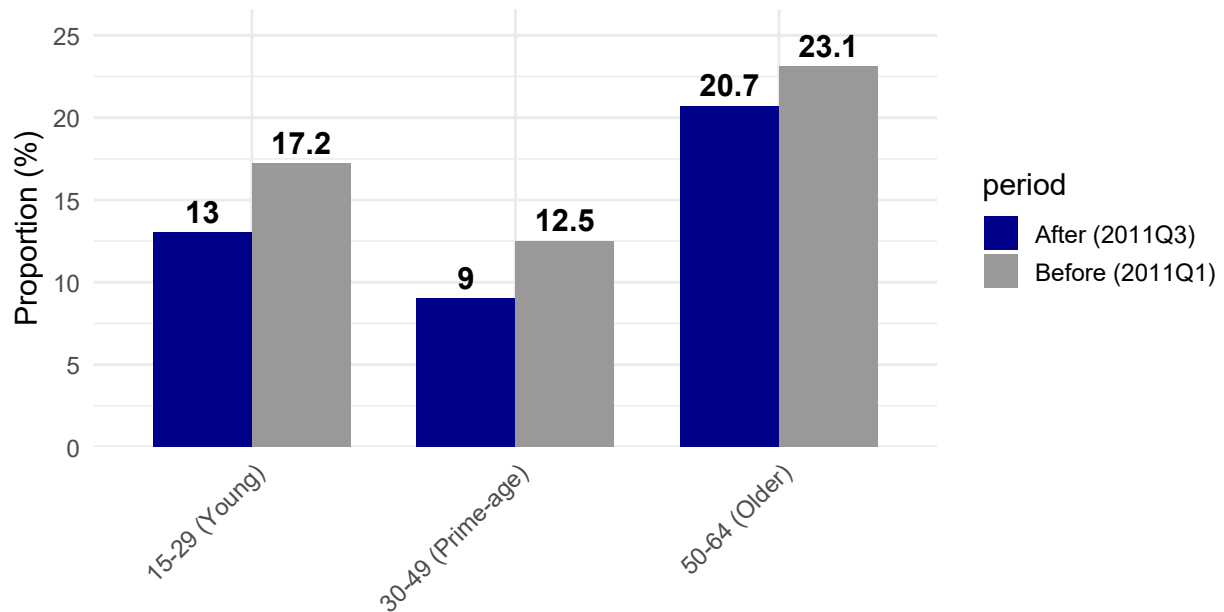
Below hk 28

Below Real HK\$28: Right Before vs Right After
2011Q1 (Before) vs 2011Q3 (After)



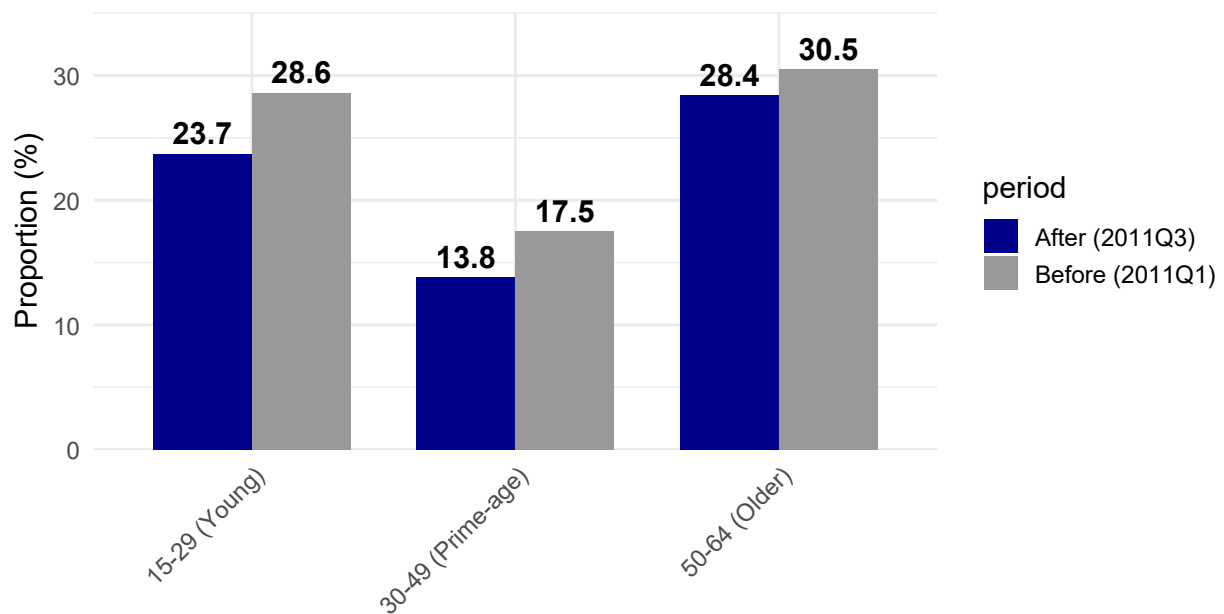
Below hk 35

Below Real HK\$35: Right Before vs Right After
2011Q1 (Before) vs 2011Q3 (After)



Below hk 40

Below Real HK\$40: Right Before vs Right After
2011Q1 (Before) vs 2011Q3 (After)



Summary table: Right before vs after by age (Below \$28)

Table 1: Right Before vs Right After: Proportion Below Real HK dollar 28 by Age Group (2011Q1 vs 2011Q3)

age_group	After (2011Q3)	Before (2011Q1)	Change (pp)	Relative Change (%)
50-64 (Older)	7.0	12.6	-5.6	-44.4
30-49 (Prime-age)	3.1	5.6	-2.5	-44.6
15-29 (Young)	4.3	6.0	-1.7	-28.3

Key Findings from Age Group Analysis:

Older workers (50-64) had the highest incidence of low wages both before and after the minimum wage. The proportion earning below HK\$28 dropped significantly from 12.6% (2011Q1) to 7.0% (2011Q3). Young workers (15-29) also benefited, with a reduction from 6.0% to 4.3%. Prime-age workers (30-49) had the lowest share of very low wages and showed a moderate improvement.

This pattern suggests that while young workers are often the focus of minimum wage discussions, older low-skilled workers in Hong Kong were actually the most vulnerable group and also gained substantially from the policy.

Causality Concern (The Most Important Point)

We cannot strongly claim “the minimum wage caused the drop” based only on before-after comparison, because other things (economic recovery, seasonal effects, etc.) could be happening at the same time.

Robustness test (very weak tho)

a simple before-after comparison is suggestive but not conclusive proof of causality. A Robustness test is a good way to strengthen your argument.

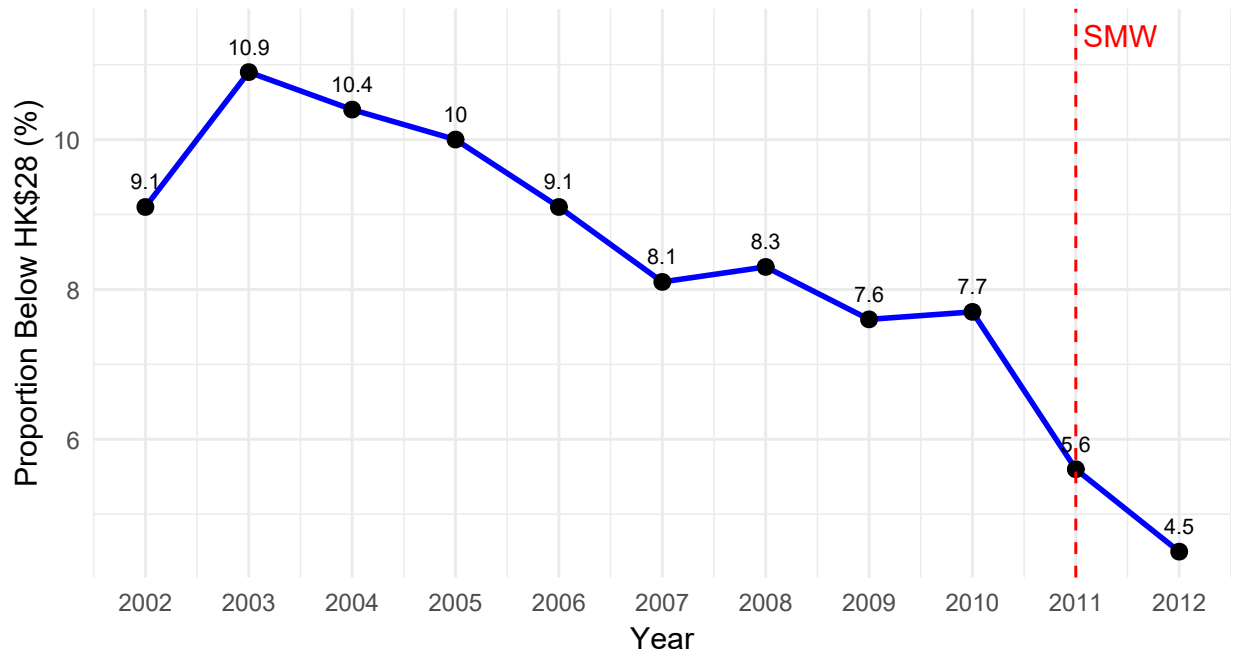
We test whether we see similar “drops” in low-wage shares in fake treatment periods (years/quarters where no minimum wage was introduced). If we only see a big drop in 2011Q2/Q3 but not in other years, it strengthens the case that the minimum wage mattered.

Proportion below HK 28 over time

```
## # A tibble: 11 x 2
##   year `Below $28`
##   <dbl>     <dbl>
## 1  2002         9.1
## 2  2003        10.9
## 3  2004        10.4
## 4  2005         10
## 5  2006         9.1
## 6  2007         8.1
## 7  2008         8.3
## 8  2009         7.6
## 9  2010         7.7
## 10 2011         5.6
## 11 2012         4.5
```

Robustness Test: Proportion Below HK\$28 Over Time

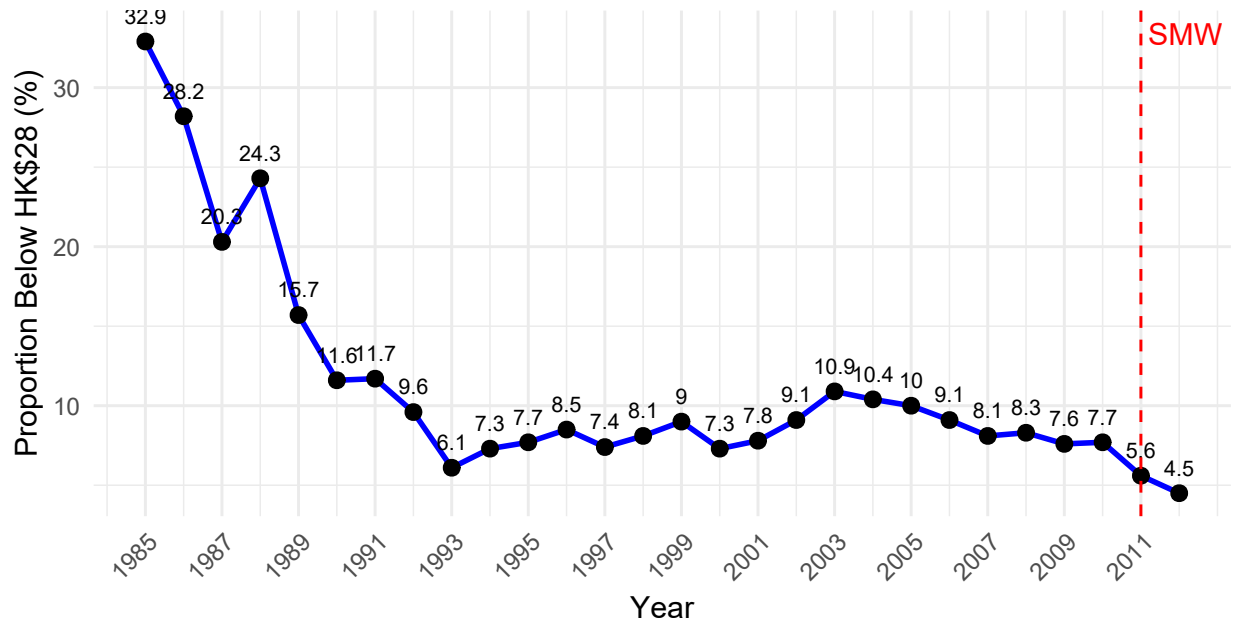
Big drop only after 2011 supports policy impact?



```
## # A tibble: 28 x 2
##   year `Below $28`
##   <dbl>     <dbl>
## 1 1985      32.9
## 2 1986      28.2
## 3 1987      20.3
## 4 1988      24.3
## 5 1989      15.7
## 6 1990      11.6
## 7 1991      11.7
## 8 1992       9.6
## 9 1993       6.1
## 10 1994       7.3
## # i 18 more rows
```

Robustness Test: Proportion Below HK\$28 Over Time (1985-2012)

A decline occurred in 2011, but similar declines happened in earlier years (e.g., 2009, 2003)



Sharp drop only in 2011, while previous years show steady/slow decline

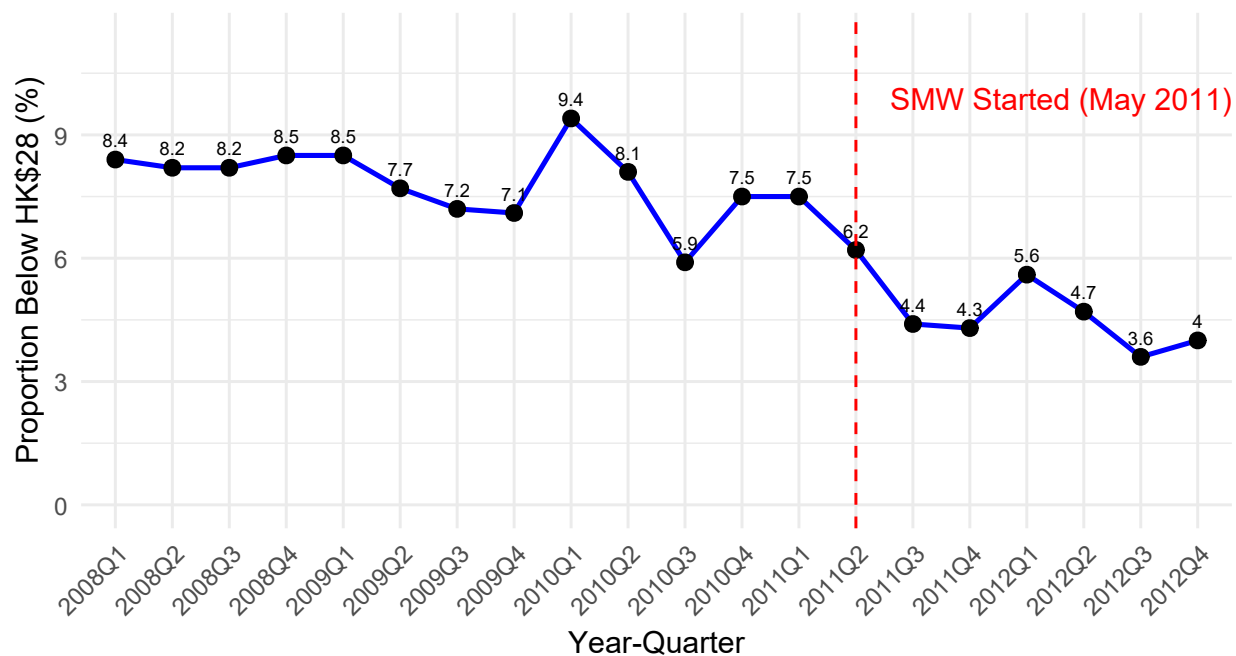
But a single yearly drop, even if large, is not fully robust proof of causality.

“what if it was economic recovery / seasonal effects / other shocks?” then boom

```
## # A tibble: 20 x 5
##   year quarter below_28     n year_quarter
##   <dbl>   <dbl>   <dbl> <int> <chr>
## 1 2008         1     8.4  5366 2008Q1
## 2 2008         2     8.2  5522 2008Q2
## 3 2008         3     8.2  5591 2008Q3
## 4 2008         4     8.5  5578 2008Q4
## 5 2009         1     8.5  5468 2009Q1
## 6 2009         2     7.7  5564 2009Q2
## 7 2009         3     7.2  5339 2009Q3
## 8 2009         4     7.1  5350 2009Q4
## 9 2010         1     9.4  5243 2010Q1
##10 2010         2     8.1  5238 2010Q2
##11 2010         3     5.9  5140 2010Q3
##12 2010         4     7.5  5423 2010Q4
##13 2011         1     7.5  5340 2011Q1
##14 2011         2     6.2  5309 2011Q2
##15 2011         3     4.4  5192 2011Q3
##16 2011         4     4.3  5401 2011Q4
##17 2012         1     5.6  5307 2012Q1
##18 2012         2     4.7  5332 2012Q2
##19 2012         3     3.6  5290 2012Q3
##20 2012         4         4  5263 2012Q4
```

Quarterly Robustness Test: Proportion Below HK\$28

Sharp drop starting in 2011Q2 / 2011Q3 supports policy impact



The quarterly data does show a drop, but it's not a massive "cliff" that screams "policy effect!" — which is very common in real-world minimum wage studies.

2011Q1 (Before): 7.5% 2011Q2 (SMW started in May): 6.2% → already dropping 2011Q3: 4.4% → sharp drop 2011Q4: 4.3% 2012 stays low (~3.6–5.6%)

Robustness tests are weak

The 2010Q3 dip (5.9%) shows the low-wage share can fluctuate significantly even without the policy

False Timing Robustness for 50-64 (the next section is all age group)

Test Every Year 1986-2010 as a Fake Treatment

1985 no 1984 data to provide drop or what not so not included, 1986 can compare to 1985 tho

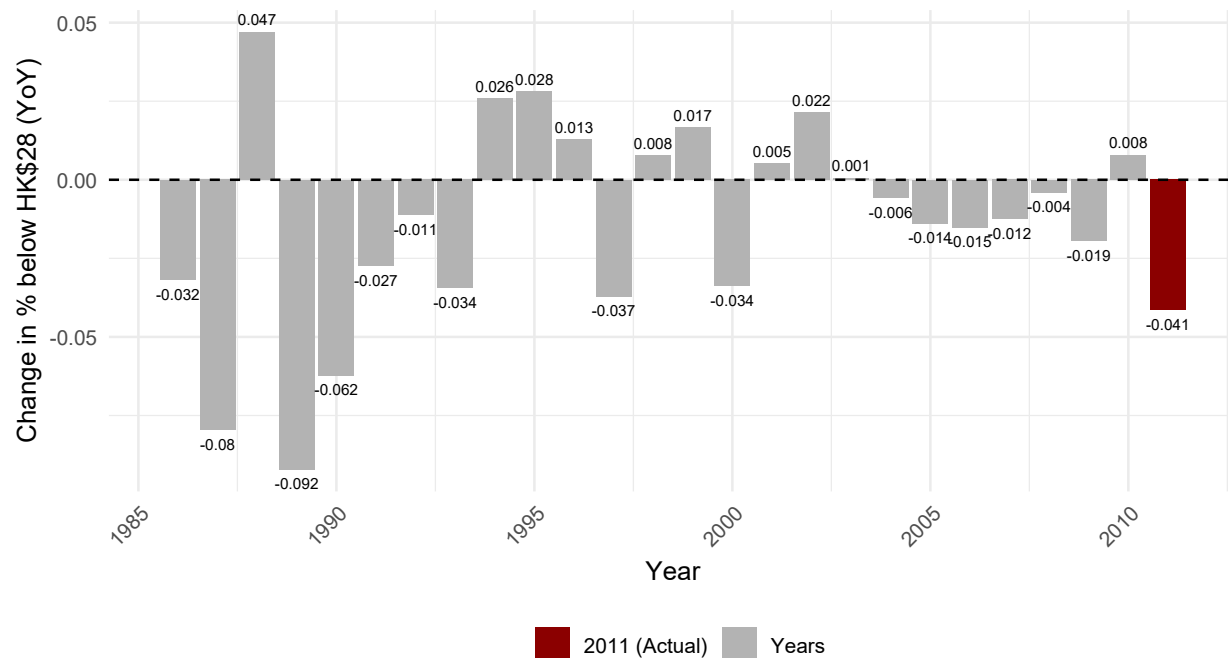
```
## # A tibble: 26 x 2
##   Year    Drop
##   <dbl>  <dbl>
## 1 1986 -0.0318
## 2 1987 -0.0797
## 3 1988  0.0470
## 4 1989 -0.0922
## 5 1990 -0.0623
## 6 1991 -0.0274
## 7 1992 -0.0113
## 8 1993 -0.0344
## 9 1994  0.0258
## 10 1995  0.0280
## # i 16 more rows
## # A tibble: 1 x 1
```

```
## max_drop
## <dbl>
## 1 -0.0922

## # A tibble: 2 x 2
##   Year Drop
##   <dbl> <dbl>
## 1 2000 -0.0337
## 2 2011 -0.0415
```

Robustness Test: Year-over-Year Drops in Low-Wage Share (50-64 Workers)

Negative = improvement. 2011 is NOT the largest drop.



Largest pre-2011 drop: 1989 = -0.0922 (9.22 percentage points)

2011 drop: -0.0415 (4.15 percentage points)

2011 is NOT the largest drop.

The drop was moderate compared to historical fluctuations

1989 saw a drop more than twice as large than 20

Cannot conclude if minimum wage cause this drop

Simple before-after or DiD with another age/education group as “control” still relies on strong assumptions (parallel trends) that we cannot properly test

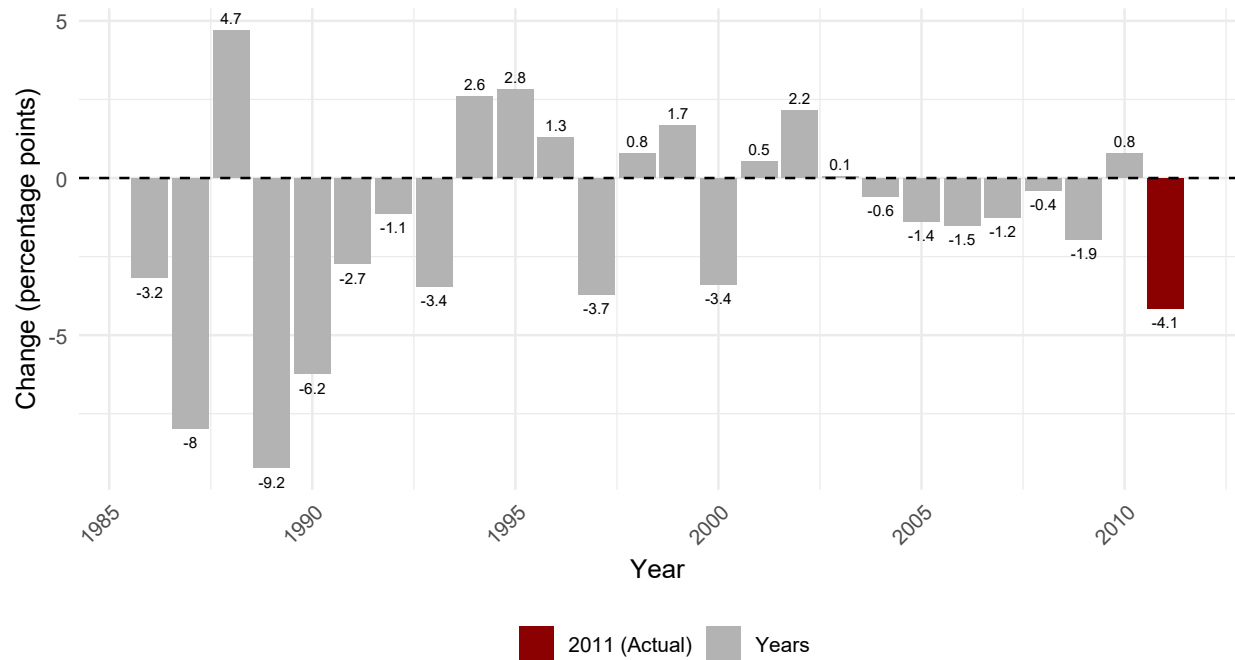
Any unobserved shock (economic recovery, industry changes, etc.) could explain part of the drop like maybe tsunami killed a bunch of low income workers so proportion dropped etc.

There was an acceleration in the decline starting around the time the minimum wage was implemented. However, the drop was not extremely dramatic, and the downward trend was already happening before 2011.

False timing Robustness for each age group:

Robustness Test: Older Workers (50-64)

Year-over-year change in % below HK\$28 | Negative = improvement



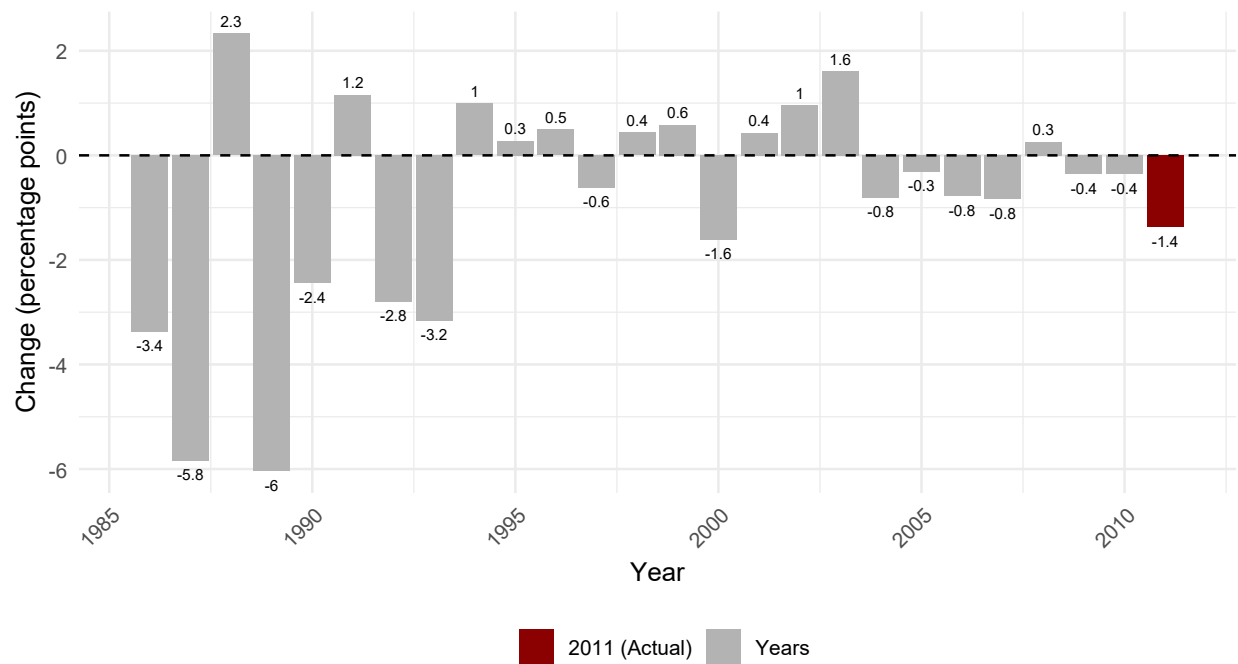
```
## # A tibble: 27 x 2
##   Year      Drop
```



```
##      <dbl>      <dbl>
## 1 1986 -0.0318
## 2 1987 -0.0797
## 3 1988  0.0470
## 4 1989 -0.0922
## 5 1990 -0.0623
## 6 1991 -0.0274
## 7 1992 -0.0113
## 8 1993 -0.0344
## 9 1994  0.0258
## 10 1995  0.0280
## # i 17 more rows
```

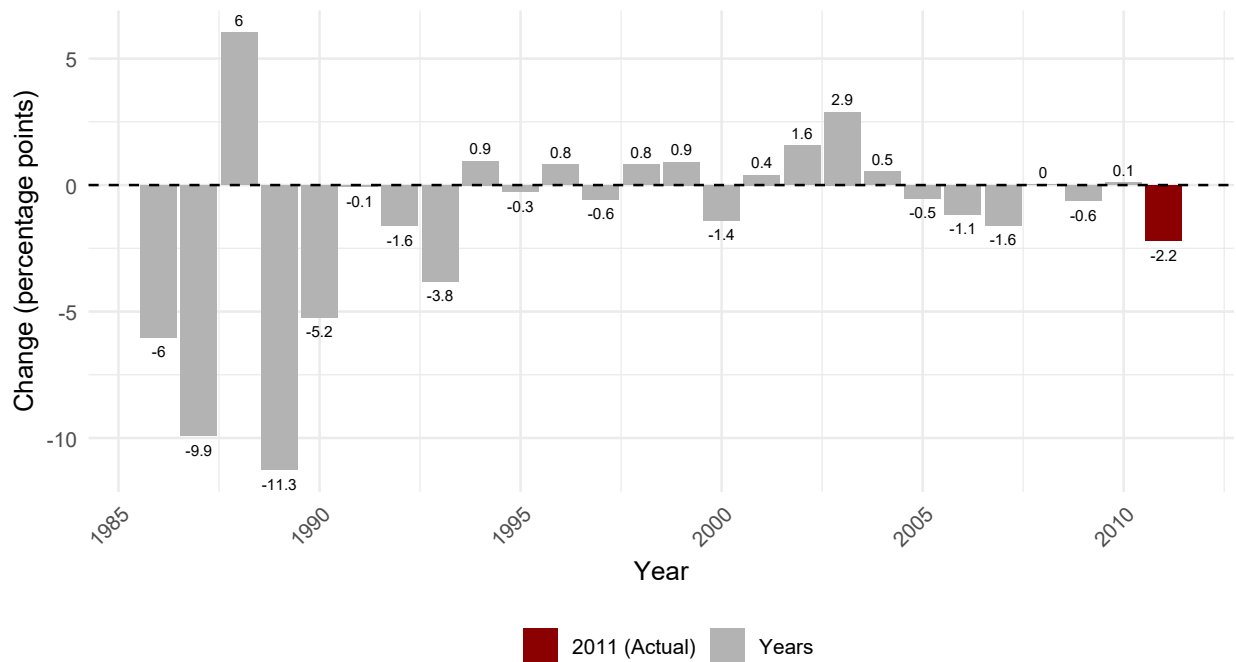
Robustness Test: Prime-Age Workers (30-49)

Year-over-year change in % below HK\$28 | Negative = improvement



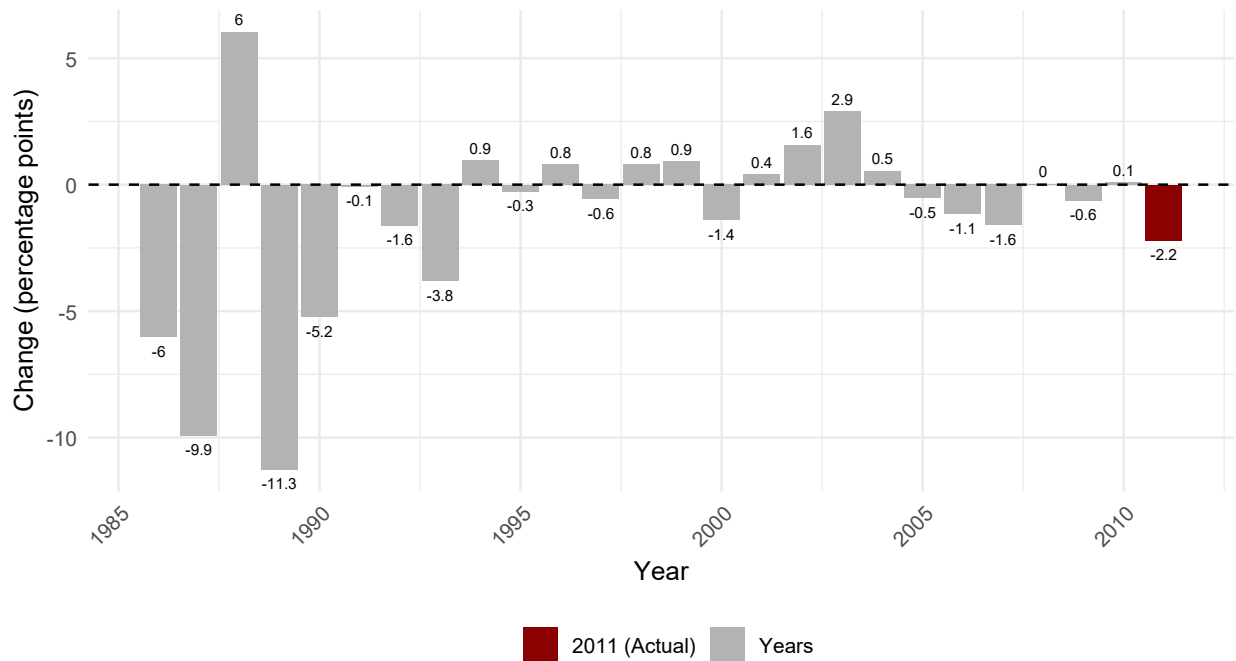
Robustness Test: Young Workers (15-29)

Year-over-year change in % below HK\$28 | Negative = improvement



Robustness Test: Young Workers (15-29)

Year-over-year change in % below HK\$28 | Negative = improvement



Age Group 2011 Drop Historical Largest Drop Verdict 50-64 (Older) ~ -0.041 ~ -0.092 (1989) 2011 is not special 30-49 (Prime) ~ -0.014 ~ -0.032 (1990) 2011 is similar to historical 15-29 (Young) ~ -0.022 ~ -0.113 (1990) 2011 is smaller than historical

Vulnerable workers (50-64) saw bigger drops in 1989 with no minimum wage

2011 was NOT special, 2011 drops are historically unremarkable

Conclusion, Frame the paper as a descriptive + suggestive analysis

introduction of the Statutory Minimum Wage in May 2011 coincided with a noticeable reduction in the proportion of workers earning very low wages, particularly among older workers (50-64) (and those with lower education levels).

While the timing and the heterogeneous pattern of changes i.e. stronger effects among the most vulnerable groups, are consistent with a policy impact, **the research design does not allow for strong causal attribution.**

Why is not strong causal?

Survey is repeated cross-sectional rather than longitudinal, and other economic factors (such as post-financial crisis recovery) were also at play during this period.

quarterly data shows that low-wage shares can fluctuate significantly even in the absence of policy changes (e.g., 2010Q2 to 2010Q3 dropped then 2010Q3 to 2010Q4 risen).

Nevertheless, the sustained lower levels of very low pay after mid-2011, especially among lower-educated and older workers, provide suggestive evidence that the minimum wage *may* have contributed to improving outcomes for Hong Kong's most vulnerable workers.

Final Answer

What happened? Low-wage employment dropped after 2011.

Was it caused by the minimum wage? Cannot conclude — similar drops happened before.

What can we say? The drop is consistent with a policy effect, but **historical fluctuations mean causal attribution is impossible.**

Is the paper worthless? No. we've documented that before-after comparisons are misleading without historical context. That's a genuine contribution.

Title of our project?

Deepseek:

Recommended Title Formula

Theme: Descriptive / Before-After / Policy Evaluation

"The Introduction of Hong Kong's Statutory Minimum Wage: Descriptive Evidence on Changes in Low-Wage Employment by Education and Age Group"

```
## [1] "Sensitivity Test Results - Older Workers (50-64) - REAL WAGES"
```

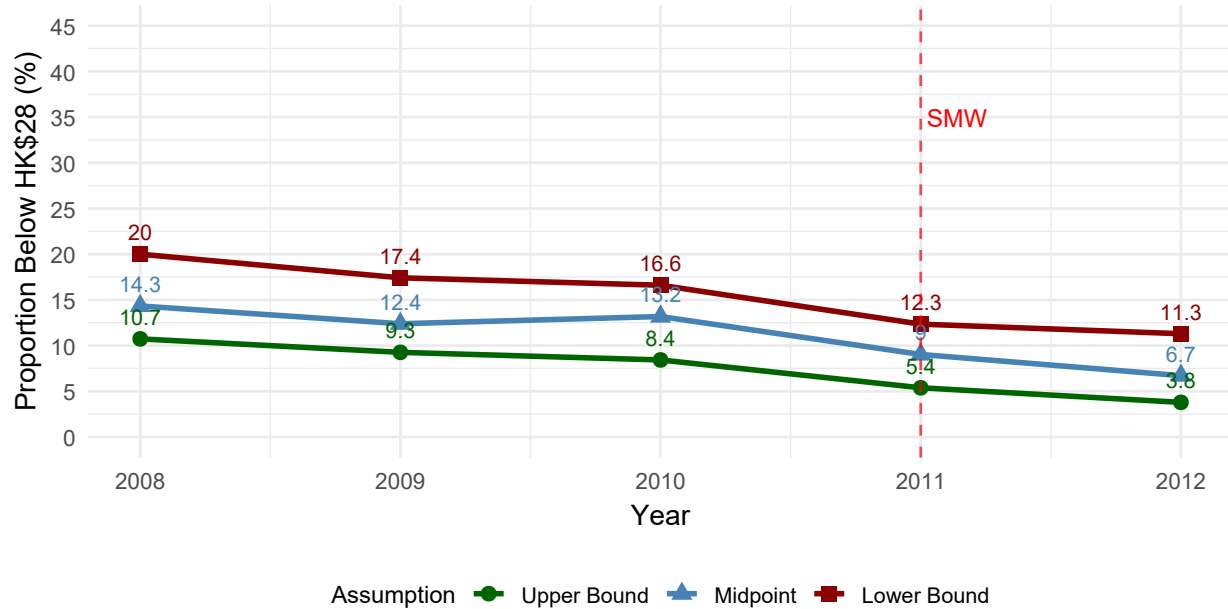
```
## # A tibble: 5 x 4
```

```
##   year `Midpoint (Real)` `Lower Bound (Real)` `Upper Bound (Real)`
##   <dbl>         <dbl>         <dbl>         <dbl>
## 1  2008          14.3          20.0          10.7
## 2  2009          12.4          17.4           9.26
## 3  2010          13.2          16.6           8.43
## 4  2011           9.03          12.3           5.39
```

5 2012 6.71 11.3 3.79

Sensitivity Test: Proportion Below HK\$28 by Earnings Assumption (Real Wages)

Older Workers (50-64) | Real wages adjusted to 2011 Q2 prices



```
## # A tibble: 572,003 x 55
##   year_quarter household_ref_no relationship age sex education
##   <chr>         <chr>          <chr>    <chr> <chr> <chr>
## 1 20001         01100020001         01      11  2    8
## 2 20001         01100170001         08      08  1    9
## 3 20001         01100350001         01      10  1    9
## 4 20001         01100490001         01      08  2    9
## 5 20001         01100490001         04      13  2    1
## 6 20001         01100560001         01      07  1    5
## 7 20001         01100560001         02      07  2    6
## 8 20001         01100650001         02      08  2    3
## 9 20001         01100760001         03      09  1    2
## 10 20001        01100760001         03      08  2    2
## # i 571,993 more rows
## # i 49 more variables: marital_status <chr>, economic_status <chr>,
## #   underemployed <chr>, work_place <chr>, industry_employed <chr>,
## #   industry_underemployed <chr>, previous_industry <chr>,
## #   occupation_employed <chr>, previous_occupation <chr>, hours_7days <dbl>,
## #   hours_main_employment <dbl>, earnings_employed <chr>,
## #   earnings_underemployed <chr>, unemployment_duration <chr>, ...
```

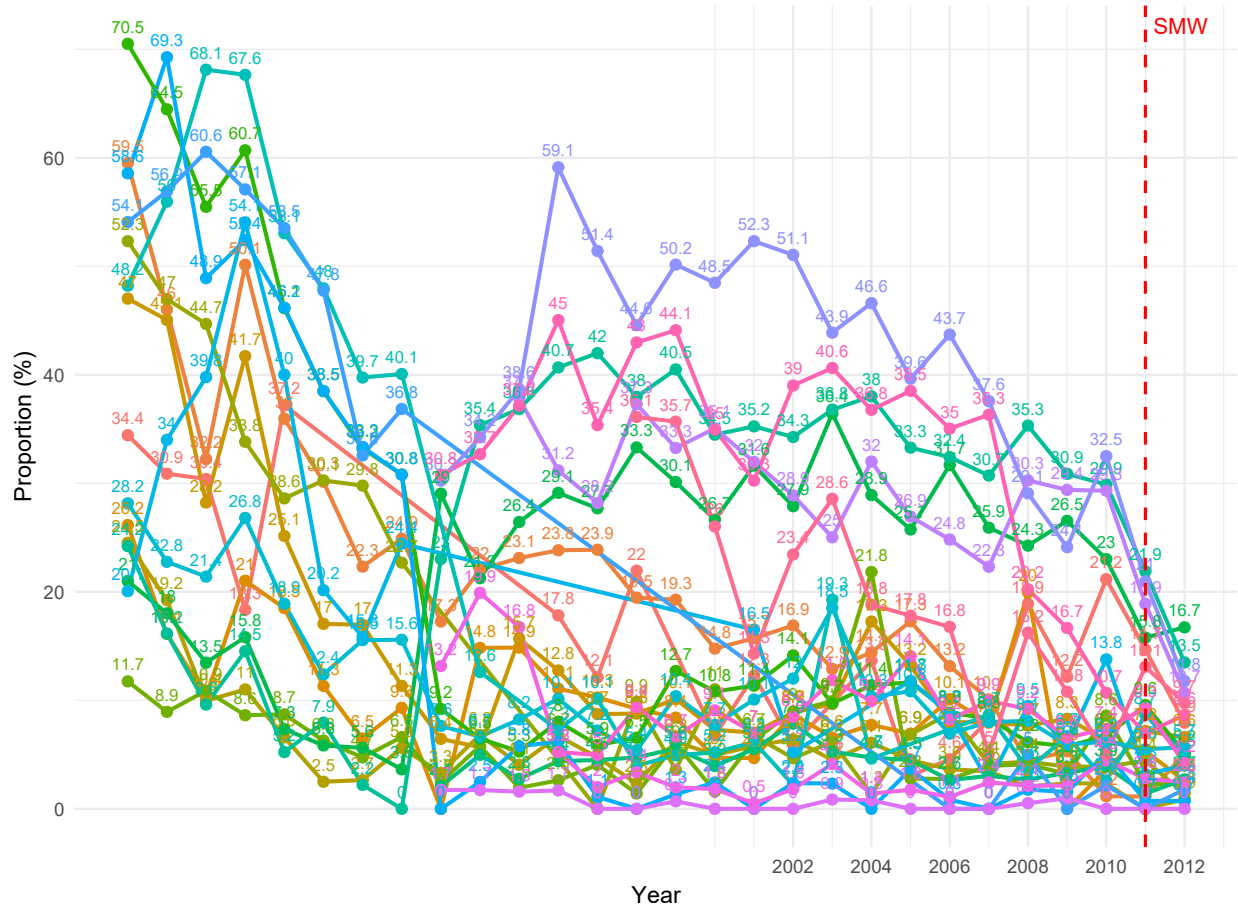
Table 2: Industry Distribution of Older Workers (50-64)

Industry Code	Industry	Weighted %	Weighted Population
02	02 Manufacturing - Clothing & Footwear	14.0	1726889.7
06	06 Construction (Services/Fitting)	14.0	1731349.1

04	04 Other Manufacturing	13.8	1706779.1
13	13 Transport & Storage	11.6	1430655.6
08	08 Wholesale Trade	8.9	1105760.2
09	09 Retail Trade	7.5	925822.8
05	05 Construction (Main Works)	7.3	901892.5
15	15 Financing	4.7	580177.2
07	07 Construction - Decoration & Maintenance	4.1	508672.1
16	16 Insurance	3.5	432476.4
12	12 Hotels / Accommodation	3.2	396454.4
03	03 Manufacturing - Paper & Printing	2.0	252091.0
11	11 Restaurants / Food Services	1.9	229579.5
14	14 Communications / Postal	1.5	190832.3
01	01 Manufacturing - Food & Beverage	1.4	173206.8
10	10 Import/Export Trade	0.6	73140.7
17	17 Real Estate	0.0	3.0
18	18 Business / Professional Services (incl. Cleaning)	0.0	2.8
19	19 Public Administration	0.0	1.4
20	20 Education	0.0	2.1
21	21 Health & Social Work	0.0	1.8
22	22 Other Community & Personal Services	0.0	3.1
23	23 Others	0.0	0.5

Proportion Earning Below Real HK\$28 by Industry

Older Workers (50-64) | Real wages in 2011 prices | Major industries only



06 Construction (Services/Fitting)	11 Restaurants / Food Services	16 Insurance
07 Construction - Decoration & Maintenance	12 Hotels / Accommodation	17 Real Estate
08 Wholesale Trade	13 Transport & Storage	18 Business / Professional Services (incl. Cleaning)
09 Retail Trade	14 Communications / Postal	19 Public Administration
10 Import/Export Trade	15 Financing	20 Education

```
## # A tibble: 13 x 4
##   year below_23 below_28 below_33
##   <dbl>   <dbl>   <dbl>   <dbl>
## 1 2000     9.80    16.7     24.7
## 2 2001     7.80    17.3     23.5
## 3 2002    10.1    19.4     27.6
## 4 2003    10.4    19.5     27.4
## 5 2004    10.1    18.9     26.7
## 6 2005     9.31    17.5     26.8
## 7 2006     7.26    16.0     23.9
## 8 2007     6.80    14.7     24.1
## 9 2008     6.69    14.3     23.3
## 10 2009     5.48    12.4     20.0
## 11 2010     5.25    13.2     20.7
```

##	12	2011	3.03	9.03	18.8
##	13	2012	2.41	6.71	15.6

Proportion Below Different Thresholds (HK\$23, HK\$28, HK\$33)

Older Workers (50-64) | Real wages adjusted to 2011 Q2 prices

